

Hospital-acquired infection surveillance in the United Kingdom and United States: contrasting architectures, metrics, incentives, and implications for prevention

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Abstract

Objective: To compare hospital-acquired infection (HAI) surveillance in the United Kingdom and United States, identify how institutional design shapes what is measured, and propose a convergent model that better supports prevention.

Design: Comparative narrative review and policy analysis of peer-reviewed literature, national protocols, technical manuals, and current public reports.

Setting: National surveillance systems in the United States and the United Kingdom, with England used as the principal UK case because health protection is devolved and England has the most extensive centrally documented mandatory programme.

Methods: Surveillance arrangements were compared across governance, case definitions, denominators, attribution rules, risk adjustment, reporting incentives, validation, post-discharge ascertainment, data automation, and public accountability. Current headline statistics were interpreted only within their originating protocols.

Results: England combines national laboratory-based surveillance of selected bloodstream infections and *Clostridioides difficile* infection with procedure-specific surgical-site surveillance and periodic point-prevalence surveys. The US National Healthcare Safety Network provides a broader modular platform focused on device-associated, procedure-associated, and laboratory-identified events, with standardized infection ratios and utilization ratios linked to state mandates and federal payment programmes. England offers strong longitudinal visibility of selected organisms across a largely unified health service, but incomplete coverage of many clinically important syndromes. The United States offers richer exposure-adjusted facility benchmarking, but greater dependence on heterogeneous local case finding, coding resources, testing behaviour, and financial incentives. Rates are not directly comparable because surveillance boundaries differ.

Conclusions: Neither architecture provides a complete account of preventable harm. A more effective model would combine sensitive automated laboratory and electronic-record detection with clinically adjudicated syndromic surveillance, transparent uncertainty, exposure denominators, post-discharge linkage, independent validation, and metrics designed for learning rather than punishment alone.

Keywords: healthcare-associated infection; hospital epidemiology; surveillance; NHSN; UKHSA; public reporting; quality measurement; infection prevention

1 Introduction

Hospital-acquired infections remain among the most visible and consequential failures of modern healthcare. They prolong admission, increase antimicrobial exposure, create avoidable disability and death, and impose costs on patients, families, hospitals, and health systems. Their significance extends beyond the infection episode itself: HAIs are both outcomes of care and signals about the reliability of clinical systems, including staffing, environmental cleaning, device management, antimicrobial stewardship, diagnostic practice, and the coordination of work across professional boundaries. Estimates of preventability vary by infection type and method, but a substantial fraction

of HAIs can be reduced through well-organised programmes that combine evidence-based practice, surveillance, feedback, and leadership accountability.[1–4]

Surveillance is therefore not merely a method of counting infections. It is an intervention that defines a field of attention. The categories selected for mandatory reporting influence which cases are investigated, which prevention teams receive resources, which clinical services are compared, and which trends become politically legible. A surveillance system also embeds judgments about causation. The label “hospital acquired” may be assigned by elapsed time after admission, exposure to a device, a recent procedure, a laboratory test, clinical signs, readmission, or expert review. Each method answers a different question. A laboratory-defined episode can be reproducible and rapid but may not distinguish infection from colonisation or reveal the responsible pathway. A clinically adjudicated event may be more meaningful for prevention but can be costly, subjective, and vulnerable to inconsistent application.[5–8]

The United Kingdom and United States offer an instructive comparison because both have mature infection-prevention traditions, national data infrastructures, public reporting, and long histories of policy concern about HAIs. Yet their health systems and surveillance architectures differ profoundly. The United Kingdom has tax-funded, territorially organised health services, with policy and public-health responsibilities devolved to England, Scotland, Wales, and Northern Ireland. England’s programme has developed around mandatory national reporting of selected bloodstream infections and *Clostridioides difficile* infection (CDI), supplemented by surgical-site infection (SSI) surveillance, outbreak systems, antimicrobial-resistance surveillance, and periodic point-prevalence surveys. The United States has a pluralistic delivery and payment system in which the Centers for Disease Control and Prevention (CDC) operates the National Healthcare Safety Network (NHSN), a modular surveillance platform used by tens of thousands of facilities. NHSN measures are closely interwoven with state reporting laws, accreditation, public comparison, and federal payment programmes.[9–12]

These differences matter because apparently simple national comparisons can be misleading. A reduction in a standardized infection ratio (SIR) in the United States is not equivalent to a reduction

in an English count per occupied bed-days, and neither is directly interchangeable with prevalence measured during a point-prevalence survey. The numerator may represent a clinically confirmed infection, a positive laboratory event, or an event meeting a device-associated definition. The denominator may be admissions, patient-days, device-days, procedures, or a modelled prediction. The exposure period and post-discharge window may differ. Testing thresholds, coding resources, surveillance intensity, and incentives also alter the probability that a case is recorded. Surveillance data are constructed observations rather than neutral reflections of an underlying burden.[13–16]

This review compares the two systems as institutional technologies for prevention. Its objectives are to describe their principal components; examine how definitions, denominators, and incentives shape observed rates; assess strengths and limitations; and identify reforms that could improve learning, comparability, and patient protection. Because health protection is devolved, the term “UK” is used cautiously. The detailed analysis uses England as the principal UK case, while recognising that Scotland, Wales, and Northern Ireland maintain separate surveillance programmes and publications.[17–20]

2 Methods and analytical framework

This paper is a comparative narrative review rather than a systematic review or pooled epidemiological analysis. Sources included national surveillance protocols and manuals; current annual reports; public descriptions of payment and reporting programmes; point-prevalence studies; and peer-reviewed research on surveillance effectiveness, public reporting, automation, validation, and the burden of HAI. Priority was given to primary national documents from the UK Health Security Agency (UKHSA), CDC, Centers for Medicare and Medicaid Services (CMS), World Health Organization (WHO), and peer-reviewed studies describing the operation or evaluation of surveillance systems. Current statistics were taken from the most recent reports available during manuscript preparation in July 2026.[8, 21–23]

The comparison was organised around ten dimensions. First, governance asks who sets

definitions, receives data, and can require participation. Second, scope concerns which organisms, syndromes, devices, procedures, and settings are represented. Third, ascertainment concerns how potential cases are detected and confirmed. Fourth, temporal attribution specifies when an event is considered present on admission or healthcare associated. Fifth, denominators and risk adjustment determine how counts become rates or comparative statistics. Sixth, reporting incentives include professional feedback, public disclosure, regulation, and financial consequences. Seventh, validation asks how reliably facilities apply definitions and how national bodies detect errors. Eighth, timeliness and workload address whether information can support action without overwhelming infection-prevention teams. Ninth, interoperability concerns linkage among laboratory, admission, procedure, pharmacy, microbiology, and community records. Tenth, equity concerns whether metrics reveal or conceal unequal exposure and outcomes among patient groups and facilities.

The term HAI is used for infections associated with healthcare broadly, while “hospital acquired” is retained where the focus is acute inpatient care or the terminology of a source. Surveillance does not always establish that an infection was caused by a particular hospital. Time-based attribution is an operational rule, not a complete causal analysis. Similarly, “community onset” does not imply absence of recent healthcare exposure. Both English and US systems increasingly use categories that recognise prior admission, procedures, long-term care, or healthcare contact, but their public headline measures often remain simpler than the underlying patient pathways.[10, 11, 24]

No attempt was made to calculate a single UK-to-US rate ratio. The relevant source populations, periods, case definitions, and denominators do not support such a calculation. Current figures are presented as surveillance snapshots to illustrate system output, not as a league table. This decision follows established guidance that public comparison requires stable definitions, appropriate risk adjustment, adequate validation, and explicit communication of uncertainty.[8, 13]

3 The epidemiological and organisational purpose of HAI surveillance

A useful surveillance system must perform several functions simultaneously. At the bedside and ward level, it should identify clusters and prevent onward transmission. At hospital level, it should provide feedback on processes and outcomes, reveal variation between units, and evaluate interventions. At regional and national levels, it should estimate burden, detect changes in organisms and resistance, direct support to poorly performing services, and inform policy. These purposes can conflict. A narrow, highly standardised indicator may be suitable for national comparison but omit many infections important to local clinicians. A sensitive local case-finding system may support prevention but be too heterogeneous for payment or public ranking. The design problem is not simply to maximise case detection; it is to align measurement with the decision that the data are intended to support.[8, 25, 26]

The historical evidence for surveillance as part of prevention is strong. The US Study on the Efficacy of Nosocomial Infection Control found that hospitals with organised surveillance and control programmes achieved substantially greater reductions in nosocomial infection than hospitals without effective programmes. Later national and multicentre initiatives demonstrated that measurement, feedback, standardised practice, and collaborative improvement could reduce device-associated and surgical infections. However, surveillance is not independently curative. Its benefit depends on whether data return to teams in a credible, timely, and actionable form and whether those teams have authority and resources to change practice.[1, 27–29]

HAI burden can be described through incidence, prevalence, attributable mortality, disability-adjusted life years, excess length of stay, antimicrobial use, and cost. European burden modelling indicates that a small group of major HAIs produces population health loss comparable with or exceeding many other communicable diseases. US economic studies similarly identify central-line-associated bloodstream infection (CLABSI), SSI, ventilator-associated pneumonia, CDI, and catheter-associated urinary tract infection (CAUTI) as costly conditions. Yet burden estimates

inherit all the limitations of their source data, especially assumptions about attributable outcomes and preventability.[30–32]

Point-prevalence surveys provide broad cross-sectional estimates that compensate for the selectivity of routine mandatory systems. In US acute-care hospitals, national multistate surveys found that approximately one in 25 inpatients had at least one HAI in 2011, with a lower prevalence observed in 2015; pneumonia, gastrointestinal infection, SSI, and other infections remained prominent. European surveys have similarly documented substantial HAI and antimicrobial-use prevalence across acute and long-term care. England’s 2023 survey found at least one HAI in 7.6% of 44,372 surveyed patients, with pneumonia and other lower respiratory-tract infection the largest group, followed by urinary-tract infection. These figures illustrate a central surveillance gap: some of the most prevalent syndromes, especially non-ventilator hospital-acquired pneumonia, are not continuously captured in national mandatory programmes with the same intensity as selected bloodstream infections or laboratory events.[15, 16, 33, 34]

Surveillance also produces behavioural effects. Public targets can mobilise leadership attention and standardise prevention, but high-stakes measures may encourage diagnostic avoidance, altered testing, disputation of cases, or concentration on reported conditions at the expense of unmeasured harm. Infection preventionists may spend large portions of their time collecting and validating data, reducing capacity for ward engagement and improvement. The appropriate response is not to abandon accountability but to design indicators that are difficult to game, transparently validated, and balanced by broader measures of burden and practice.[7, 13, 14, 35]

4 Surveillance architecture in the United Kingdom

4.1 Devolution and the use of England as the principal case

There is no single operational UK surveillance database that can be described without qualification. England, Scotland, Wales, and Northern Ireland have separate health services, public-health agencies, reporting pathways, and publication series. Common scientific principles and professional networks

promote broad alignment, and all four nations undertake surveillance of major HAIs, antimicrobial resistance, and outbreaks. Nevertheless, mandatory categories, time series, denominators, dashboards, and governance arrangements are not identical. A paper that equates England with the whole United Kingdom would therefore overstate institutional uniformity.[17–19]

England is used here as the principal UK case for three reasons. It contains most of the UK population and hospital activity; its mandatory programme has a long, documented history; and its current protocols and trust-level outputs permit detailed comparison with NHSN. Scotland has conducted national prevalence surveys and publishes healthcare-associated infection reports, while Wales and Northern Ireland operate their own organism- and syndrome-based surveillance. These systems are important comparators in their own right, but a full four-nation analysis would require separate treatment of changing definitions and local governance.[17–20]

4.2 Mandatory organism-based surveillance in England

England’s national mandatory programme emerged from intense concern about methicillin-resistant *Staphylococcus aureus* (MRSA) bacteraemia. Mandatory reporting established a national count and rate for each acute trust and became linked to performance management. The programme subsequently expanded to include methicillin-susceptible *S. aureus* (MSSA) bacteraemia, *Escherichia coli* bacteraemia, other Gram-negative bacteraemias, and CDI. Data are submitted through a national data capture system using laboratory episodes combined with patient, admission, location, and attribution fields. This architecture gives England a consistent longitudinal view of selected organisms across the National Health Service.[9, 10, 26]

The strength of laboratory-based mandatory reporting is ascertainment at scale. Positive blood cultures and toxin or molecular test results are routinely generated, electronically identifiable, and less dependent than some clinical syndromes on narrative interpretation. National requirements can achieve broad institutional coverage, and publication at trust and commissioning levels creates visibility for boards, regulators, patients, and local infection-control teams. The marked decline in MRSA bacteraemia after the expansion of national control efforts demonstrates how surveillance,

197 targets, screening, antimicrobial policy, cleanliness, and wider infection-prevention measures can
198 operate as a coordinated programme, although attribution of the decline to any single component is
199 not possible.[9, 36, 37]

200 The same design creates limitations. A positive blood culture represents the end point of
201 multiple possible pathways: vascular-device infection, urinary infection, SSI, pneumonia, skin
202 and soft-tissue infection, or community disease. Organism surveillance can identify burden and
203 location but may not by itself specify the preventable hospital process. Conversely, many clinically
204 important HAIs never produce a positive blood culture. National attention can thus be concentrated
205 on organisms that are easy to count rather than syndromes that account for the largest prevalence. *E.*
206 *coli* bacteraemia illustrates the boundary problem because a large majority of cases are not classified
207 as hospital onset, yet many community-onset cases occur in patients with recent healthcare exposure,
208 urinary catheters, prior antimicrobial treatment, or recurrent admission.[10, 21, 26]

209 Temporal attribution in the English protocol is explicit but indicator specific. For bacteraemia,
210 a case is generally categorised as hospital onset when the specimen date is on or after the third
211 day of admission, with the admission date counted as day one. CDI uses a later hospital-onset
212 threshold, generally specimen date on or after the fourth day. Additional categories describe
213 community-onset cases with recent trust exposure, making the dataset more informative than a
214 binary hospital/community split. These rules improve reproducibility, but elapsed time remains a
215 proxy. Incubation periods differ; prior care may matter; and an infection detected early can still be
216 healthcare associated.[10]

217 The 2024–2025 English report illustrates the output of this system. It recorded 43,639 *E.*
218 *coli* bacteraemia cases, of which 8,397 were hospital onset; 1,064 MRSA bacteraemias, including
219 367 hospital-onset cases; 13,799 MSSA bacteraemias; and 18,970 CDI cases, of which 8,430 were
220 hospital onset. These figures are useful for trend analysis within the English definitions. They
221 should not be interpreted as the complete national burden of HAI, nor compared directly with NHSN
222 events, because they represent different mixtures of all-origin and hospital-onset episodes and do
223 not use the same exposure denominators or risk models.[21]

4.3 Surgical-site infection surveillance

England's Surgical Site Infection Surveillance Service provides procedure-specific surveillance across a defined set of surgical categories. Participation is mandatory for at least one quarter each year in selected orthopaedic categories, while surveillance in other categories is voluntary. Hospitals submit operation-level risk factors and infections detected during the inpatient stay or on readmission, enabling calculation of procedure-specific rates and comparison with national benchmarks. The 2024–2025 report included 146,411 operations submitted by 190 NHS hospitals and additional independent-sector treatment centres, with 1,250 SSIs identified through inpatient or readmission surveillance. Most submitted operations were in the mandatory orthopaedic categories.[22]

Procedure-level surveillance is more closely linked to an actionable pathway than organism counts. It can direct attention to prophylaxis timing, skin preparation, theatre practice, normothermia, glycaemic control, wound management, and post-operative care. Yet the English scheme's mandatory scope is narrow relative to the range of procedures performed. Voluntary participation produces selective coverage, and surveillance intensity varies. Studies of English hospitals have identified heterogeneity in methods, participation, and post-discharge detection, as well as a mismatch between national surveillance coverage and the burden or priority assigned to some procedures.[38–40]

Post-discharge ascertainment is a persistent challenge. Shorter length of stay moves a growing portion of the SSI risk period into the community. Infections may be treated by primary care, emergency departments, outpatient clinics, or another hospital without returning to the operative centre. Readmission-based surveillance is specific but misses less severe infections and cases managed elsewhere. Patient questionnaires increase sensitivity but can over-identify superficial wounds unless clinically verified. Linked records and microbiology data offer a route to more complete follow-up, but national implementation requires interoperable identifiers, agreed algorithms, and governance for data sharing.[40–42]

4.4 Point-prevalence, antimicrobial-use, and supplementary systems

Periodic point-prevalence surveys broaden the field beyond continuously mandated indicators. England's 2023 survey used European-compatible methods to examine HAIs, antimicrobial use, and stewardship. Among 44,372 patients, 3,359 had at least one HAI, corresponding to a prevalence of 7.6%. Pneumonia and lower respiratory-tract infection accounted for 29.6% of recorded HAIs, urinary-tract infection for 17.5%, and SSI for 8.9%. A validation exercise estimated high specificity but imperfect sensitivity, with an adjusted prevalence slightly above the observed value. Such validation is valuable because it makes under-ascertainment visible rather than presenting surveillance as error free.[34]

Point-prevalence surveys provide breadth but not continuous feedback. They are sensitive to season, case mix, bed occupancy, and the prolonged stay of infected patients, which can make prevalence disproportionately reflect infections associated with long admissions. Repetition at intervals can show structural changes, but it cannot substitute for incidence surveillance of rapidly changing risks. Its best role is complementary: identifying blind spots in routine reporting, measuring antimicrobial exposure, and testing whether national mandatory indicators reflect the syndromes causing most current harm.[8, 16, 43]

England also uses outbreak reporting, laboratory reference services, genomic surveillance, antimicrobial-resistance systems, and local infection-prevention datasets. Whole-genome sequencing has demonstrated that hospital attribution based only on time and location can oversimplify transmission, as genetically related CDI may arise from sources not captured by conventional ward contact. Genomics can distinguish transmission clusters and support outbreak control, but it does not replace clinical or denominator data. Its greatest value comes when integrated with patient movement, exposure, microbiology, and epidemiological investigation.[24, 36]

4.5 Strengths and weaknesses of the English model

The English architecture has four major strengths. First, national mandates within a largely unified health service provide broad coverage and long time series. Second, laboratory events are

comparatively amenable to automated extraction and consistent episode construction. Third, public trust-level reporting creates a clear line of accountability. Fourth, the combination of mandatory data, prevalence surveys, and procedure surveillance allows different views of the problem. These features supported sustained national action on MRSA and CDI and provide policy makers with stable signals about selected pathogens.[9, 21, 36]

Its principal weakness is selectivity. The routinely prominent indicators do not constitute a comprehensive incidence system for all major HAIs. Hospital-acquired pneumonia, non-catheter urinary infection, many device-related events, and a wide range of SSIs are incompletely represented in mandatory national reporting. Organism-centred surveillance can separate cases by onset and prior exposure but often lacks the device-days, procedure-specific expected events, or patient-level clinical adjudication required to compare causal pathways. The consequence is a gap between what is most visible to national accountability and what may consume the greatest local prevention effort.[26, 34, 39]

A second weakness is dependence on manual enhancement. Laboratory detection may be automated, but attribution, data completion, de-duplication, review, and root-cause investigation require professional time. Infection-prevention teams must reconcile records across microbiology, admissions, wards, procedures, and community contacts. This work produces valuable local knowledge, yet repeated manual transcription can introduce delay and error. Recent English research suggests that centrally implemented algorithms using linked electronic records can achieve useful case ascertainment, but automation must be evaluated against clinically meaningful reference standards and should expose uncertain cases for review rather than silently converting probabilities into facts.[41, 42, 44]

5 Surveillance architecture in the United States

5.1 The National Healthcare Safety Network

NHSN is the central technical infrastructure for US HAI surveillance. It grew from earlier CDC nosocomial infection systems into a web-based network with components for patient safety, healthcare personnel safety, antimicrobial use and resistance, long-term care, dialysis, and other settings. The Patient Safety Component contains modules for device-associated events, procedure-associated events, multidrug-resistant organisms and CDI, neonatal and maternal measures, and additional safety events. More than 38,000 active facilities contributed to the national 2024 progress report, demonstrating a scale that extends far beyond a voluntary research network.[5, 11, 23]

The core NHSN approach is definitional standardisation. Detailed protocols specify eligible locations, surveillance periods, signs and symptoms, laboratory criteria, repeat infection timeframes, secondary bloodstream attribution periods, device association, and exclusions. In general acute-care HAI attribution, an infection is considered present on admission when the date of event falls within the defined present-on-admission window around admission; an event on or after the third calendar day of admission is healthcare associated. Individual modules can use additional rules. Laboratory-identified CDI and MRSA bloodstream-event reporting, for example, applies specimen-date and location rules that are not identical to clinically adjudicated infection definitions.[6, 11]

This detailed rule set permits surveillance across facilities that differ in ownership, information systems, clinical organisation, and payment. It also creates a substantial cognitive and administrative burden. Infection preventionists must apply changing criteria consistently, obtain denominator data, resolve event attribution, and document exclusions. Definitions designed for comparability may diverge from clinicians' judgments about whether infection is present or caused by a device. The NHSN event is a surveillance construct, not a substitute for diagnosis or a determination of negligence.[5, 7, 11]

5.2 Device- and procedure-associated surveillance

A defining strength of NHSN is its use of exposure denominators. CLABSI rates can be related to central-line days, CAUTI to urinary-catheter days, and ventilator-associated events to ventilator days. Procedure-associated surveillance uses the number and characteristics of eligible operations. These denominators make the measure more closely responsive to the opportunity for harm than a crude count per admission. They also reveal device utilisation as a prevention target: avoiding unnecessary lines or catheters can reduce both exposure and infection.[11, 45]

The distinction between outcome and utilisation is formalised through the SIR and standardized utilization ratio (SUR). The SIR compares observed infections with the number predicted from a national baseline model after adjustment for included risk factors. An SIR below one indicates fewer observed events than predicted, while an SIR above one indicates more. The SUR applies an analogous observed-to-predicted framework to device use. These measures allow facilities with different case mix to be compared more fairly than through crude rates, although fairness depends on model specification, baseline choice, data quality, and whether relevant risk factors are included.[23, 45]

NHSN SSI surveillance covers numerous inpatient operative-procedure categories, with risk adjustment based on factors defined for each procedure. CMS reporting requirements focus on selected measures, including colon surgery and abdominal hysterectomy in acute-care hospitals, while facilities may conduct broader surveillance for local or state purposes. The modular structure is therefore both expansive and selective: the technical platform can support many procedures, but the subset that receives sustained attention is shaped by mandates, payment, and local priorities.[11, 46, 47]

Device-associated definitions have prompted longstanding debate. A bloodstream infection in a patient with a central line may be categorised as CLABSI without definitive proof that the line caused it, because the surveillance definition is intentionally designed for consistent population measurement rather than individual causal adjudication. Conversely, exclusions and secondary-source attribution can remove cases that clinicians consider device related. These tensions are not

defects that can be eliminated by a perfect definition; they reflect the trade-off between sensitivity, specificity, reproducibility, and preventability. They become problematic when users interpret a surveillance label as an unqualified causal verdict.[5, 11]

5.3 Laboratory-identified events

NHSN Laboratory-identified (LabID) event reporting was designed to reduce some of the subjectivity and workload of traditional clinical surveillance for CDI and selected multidrug-resistant organisms. It relies primarily on laboratory test results, specimen timing, location, and rules for recurrent or duplicate events. LabID data can be generated more consistently from laboratory and admission systems and are well suited to broad mandatory reporting. However, positive tests are influenced by diagnostic stewardship, test sensitivity, repeat-testing practices, and the prevalence of colonisation. Comparative research has shown that LabID and traditional surveillance can produce materially different CDI classifications.[6, 11]

The attraction of LabID reporting resembles England’s organism-based programme: both obtain scale and reproducibility by starting with microbiology. The institutional context differs. In NHSN, laboratory events sit within a larger architecture containing device and procedure modules, risk-adjusted national models, and payer-linked measures. In England, organism reporting is the dominant continuous mandatory national view and is supplemented by a narrower procedure system. This difference affects the questions that national dashboards naturally answer. England readily shows the national and trust burden of selected bacteraemias; NHSN more readily compares facility performance for defined device and procedure pathways.[21, 23]

5.4 Public reporting, regulation, and payment

US HAI surveillance is inseparable from its incentive environment. Many facilities report because of state laws, federal quality programmes, or participation conditions, while others submit additional modules voluntarily. CMS’s Hospital Inpatient Quality Reporting Program requires specified HAI data, and the Hospital-Acquired Condition Reduction Program reduces Medicare payment by 1% for

hospitals whose performance falls in the worst-performing quartile on the programme's total score. Public reporting on federal comparison platforms adds reputational consequences.[12, 23, 46]

Financial linkage can secure executive attention, investment, and timely submission. It can also intensify disputes about definitions and create incentives to modify testing or documentation. Hospitals serving highly complex or disadvantaged populations may be concerned that risk adjustment does not capture all structural differences. Because penalties operate at hospital level, they may combine genuine prevention performance with case mix, surveillance resources, diagnostic practice, and statistical instability in smaller denominators. Validation and transparent methods are therefore essential to protect both patients and measured organisations.[13, 14, 45]

The 2024 national progress report shows substantial improvement in several acute-care indicators compared with 2023: 9% lower CLABSI, 10% lower CAUTI, 2% lower ventilator-associated events, 4% lower colon-surgery SSI, 7% lower hospital-onset MRSA bacteraemia, and 11% lower hospital-onset CDI. Abdominal-hysterectomy SSI increased by 8%. These results demonstrate the value of a stable national platform for detecting divergent trends rather than assuming that all HAIs move together. They should nevertheless be read as changes in NHSN-defined measures among reporting facilities, not as a complete census of every hospital-acquired infection in the United States.[23]

5.5 Validation and data heterogeneity

NHSN provides common definitions, analytic methods, training resources, and automated checks, but primary data collection remains distributed across thousands of organisations. Hospitals vary in electronic-record capability, infection-prevention staffing, laboratory practice, coding, and experience. State health departments also vary in validation intensity. CDC has cautioned that jurisdictions conducting more vigorous validation can identify additional events and therefore appear to have higher SIRs than jurisdictions with less intensive review. This creates a paradox: better ascertainment can initially look like worse performance.[13, 23]

The same paradox applies within hospitals. A well-resourced infection-prevention pro-

gramme may find more cases than a poorly resourced programme, especially for clinically adjudicated syndromes. High-stakes comparison can therefore reward under-detection unless validation is sufficiently strong. Automated extraction can improve consistency for structured elements, but it can also propagate local data errors or fail when documentation patterns change. The most credible approach combines national specifications, local clinical review, independent audits, and statistical methods that indicate uncertainty around low-volume estimates.[7, 8, 44]

5.6 Pandemic stress and resilience

The COVID-19 pandemic exposed the dependence of HAI prevention on basic system capacity. US NHSN data documented significant increases in several HAIs during 2020, associated with high occupancy, staffing disruption, altered device use, supply pressures, and the redirection of infection-prevention resources. Subsequent improvement demonstrates recovery but does not erase the lesson that surveillance indicators are partly measures of organisational resilience. A hospital can possess technically sound protocols yet experience rising infection when staffing, isolation capacity, cleaning, and routine device review deteriorate under pressure.[23, 48]

The pandemic also accelerated electronic reporting and highlighted the need for near-real-time data. Annual retrospective indicators are insufficient during rapidly evolving threats. Both countries need architectures capable of switching between routine quality surveillance and outbreak intelligence without creating parallel, incompatible reporting systems. This requires common identifiers, laboratory interfaces, admission and movement data, and governance arrangements established before a crisis.[8, 42]

6 Direct comparison of surveillance design

Table 1 summarises the principal contrast. The English model is relatively centralised, pathogen oriented, and embedded in a national health service. The US model is platform oriented, modular, and embedded in a pluralistic system of facility reporting, state requirements, and federal payment.

Neither description is absolute: England has procedure surveillance and local device data, while NHSN includes organism-based LabID events. The distinction concerns institutional emphasis rather than mutually exclusive categories.

Table 1: Principal characteristics of HAI surveillance in England and the United States

Dimension	England (principal UK case)	United States
Governance	National specifications and mandatory reporting within the NHS; separate systems in devolved UK nations	CDC NHSN technical platform; reporting driven by federal programmes, state laws, and facility choice
Core emphasis	Selected bacteraemias and CDI, onset and prior-exposure categories; SSI service; periodic prevalence surveys	Device-associated, procedure-associated, LabID, antimicrobial-use/resistance, and other modular events
Primary units	Organism episodes, trust attribution, procedures, surveyed patients	Facility locations, device-days, procedures, laboratory events, predicted events
Comparison metrics	Counts and rates using population, bed-day, or admission denominators; procedure-specific SSI rates	SIRs, SURs, pooled means, percentiles, and event rates with national risk models
Incentives	Public reporting, national targets, regulatory and managerial accountability	Public reporting, state mandates, accreditation and federal payment consequences
Major strength	Longitudinal national visibility for selected pathogens across a comparatively integrated service	Rich exposure-adjusted benchmarking for devices and procedures across many facility types
Major limitation	Incomplete continuous national coverage of common syndromes and many procedures	Heterogeneous local ascertainment and strong sensitivity to reporting resources, testing, and incentives

6.1 Case definitions and attribution

Both countries use time after admission as a major attribution device, and both recognise that timing is not identical to causation. For many bloodstream events, England’s hospital-onset threshold begins with a specimen on or after admission day three. NHSN’s general HAI framework similarly places the date of event on or after the third calendar day outside the present-on-admission period. However, module-specific details differ, including the use of symptoms, cultures, device association, repeat-event windows, and CDI LabID timing. Superficially similar “day three” or “day four” labels

cannot be assumed to produce equivalent numerators.[10, 11]

England’s enhanced prior-exposure categories are particularly valuable for a healthcare system in which many infections cross the hospital-community boundary. They can distinguish a community-onset specimen in a patient recently discharged from the reporting trust from an episode without that exposure. NHSN also uses location and prior-event rules, but national public measures often foreground hospital-onset and facility-attributed outcomes. Both systems could better represent longitudinal pathways across acute hospitals, outpatient care, long-term care, dialysis, and home devices.[10, 11]

Clinical criteria create a different form of comparability. NHSN’s device and SSI definitions provide detailed symptom and laboratory combinations, whereas England’s national bacteraemia indicators start from laboratory episodes and apply onset attribution. The former offers greater pathway specificity but requires more interpretation; the latter offers greater automation and national consistency but less clinical granularity. An optimal system would not choose one method universally. It would use laboratory events as a sensitive detection layer and add clinical or pathway adjudication where causal interpretation is required.[6–8]

6.2 Denominators, risk adjustment, and the meaning of performance

The denominator determines the implied prevention question. Cases per population ask about public-health burden. Cases per admission or occupied bed-days ask about hospital burden relative to activity. Device infections per device-days ask about risk while exposed. SSIs per operation ask about procedural risk. An SIR asks whether observed events differ from those predicted by a national model. These are complementary rather than competing measures, but public presentation often compresses them into a single notion of “rate.”[8, 45]

NHSN’s SIR and SUR are major analytic advantages for facility benchmarking. They accommodate selected differences in procedure, location, patient, and facility characteristics and convert heterogeneous denominators into interpretable observed-to-predicted ratios. Yet model-based adjustment can conceal important assumptions. A hospital’s expected number depends on the

baseline period, variable definitions, model coefficients, and exclusions. Rebaselining can alter the apparent level of performance even when raw events are unchanged. Stakeholders need access to observed counts, denominators, predicted values, confidence intervals, and model documentation rather than a single colour-coded score.[23, 45]

England's simpler rates can be more transparent but may be less able to separate case mix from preventable variation. Trust-level bacteraemia counts per bed-days or population are readily understood, yet referral centres, specialist services, and differences in community burden can complicate comparison. Procedure surveillance includes risk factors, but national mandatory coverage remains concentrated in selected orthopaedic categories. England could adopt more exposure-adjusted metrics without reproducing every feature of US payment measurement; conversely, US reporting could make greater use of simple burden measures alongside SIRs.[14, 21, 22]

6.3 Breadth versus depth

The English system has depth for a small number of organisms: long time series, national coverage, patient-level enhancement, and explicit healthcare-exposure attribution. NHSN has breadth across modules and settings but participation and mandatory scope vary by measure and jurisdiction. The policy trade-off is between maintaining a small set of stable, high-completeness indicators and expanding to conditions that require more complex ascertainment. Expansion without automation risks overwhelming infection-prevention teams; excessive narrowing risks making unmeasured infection invisible.[11, 26, 35]

The contrast is clearest for pneumonia. English prevalence data and US point-prevalence research identify pneumonia as a major HAI category, but routine national performance systems have historically focused on ventilated patients or ventilator-associated event constructs rather than all non-ventilator hospital-acquired pneumonia. This leaves a large burden with limited standardised continuous surveillance. The omission reflects genuine measurement difficulty: diagnosis is clinically contested, imaging and microbiology are imperfect, aspiration is common, and attribution

is uncertain. Yet difficulty should motivate staged algorithm development rather than permanent invisibility.[8, 15, 34]

6.4 Public accountability and unintended consequences

England's targets and public trust-level statistics and the US linkage of NHSN measures to public reporting and payment both demonstrate that surveillance is a governance instrument. Strong accountability can turn infection prevention into a board-level priority. It can also narrow organisational attention to measured outcomes and generate defensive behaviour. The difference is one of degree and mechanism: US measures are more directly embedded in payer programmes, while English measures operate through national management, regulatory scrutiny, and public-service accountability.[9, 12, 14]

The ethical question is not whether public reporting should occur, but how to make it informative and fair. Patients have a legitimate interest in knowing whether organisations recognise and reduce avoidable harm. Facilities also need confidence that published differences reflect performance rather than surveillance intensity or unmeasured case mix. A fair system should publish data quality indicators, validation coverage, uncertainty intervals, and explanations of definition changes. It should avoid turning small numerical differences into categorical judgments and should reward improvement as well as absolute status.[13, 23]

6.5 Current surveillance snapshots

Table 2 presents selected recent outputs. It intentionally places the protocol context beside each number. The figures should not be used to infer that one country has more or less HAI than the other. England's organism counts, English point prevalence, US patient-prevalence shorthand, and NHSN year-on-year SIR changes are measurements of different constructs.

Table 2: Selected recent surveillance outputs and their interpretation

System and period	Reported output	What it measures	Key limitation for comparison
England mandatory HCAI surveillance, 2024–2025	43,639 <i>E. coli</i> bacteraemias (8,397 hospital onset); 1,064 MRSA bacteraemias (367 hospital onset); 18,970 CDI cases (8,430 hospital onset)	National laboratory episodes with organism-specific onset and exposure attribution	Not a complete set of clinical HAIs; mixture of all-origin and hospital-onset counts
England point-prevalence survey, 2023	7.6% of 44,372 patients had at least one HAI	Cross-sectional prevalence of multiple syndromes using survey definitions	Prevalence is affected by length of stay, season, and one-period ascertainment
England SSI surveillance, 2024–2025	146,411 operations and 1,250 inpatient/readmission SSIs	Procedure surveillance, predominantly mandatory orthopaedic categories	Voluntary coverage outside selected categories and incomplete post-discharge capture
United States national shorthand	Approximately 1 in 31 hospital patients has at least one HAI on a given day	Burden estimate derived from prevalence-survey evidence	Not an annual incidence rate or direct equivalent of English mandatory counts
US NHSN acute-care trends, 2024 versus 2023	Reductions in CLABSI, CAUTI, VAE, colon SSI, MRSA, and CDI; increase in abdominal-hysterectomy SSI	Change in NHSN-defined national measures among reporting facilities	Depends on NHSN definitions, modelled expectations, participation, testing, and validation

7 What each country can learn from the other

7.1 Lessons for the United Kingdom

The first US lesson is the value of exposure denominators. Device-days and procedure-specific predicted events connect surveillance to modifiable clinical pathways. England's organism surveillance could be complemented by a nationally supported core set of device-associated and procedure-associated measures, selected according to burden and actionability rather than ease of counting alone. This should not require immediate universal manual surveillance. A staged programme could begin with units where electronic documentation of devices and procedures is reliable, publish data quality alongside outcomes, and expand only after validation.[11, 26, 45]

The second lesson is modular infrastructure. NHSN gives facilities a common technical and definitional framework across multiple domains. England's programmes are nationally coordinated, but data can remain divided among organism reporting, SSI surveillance, antimicrobial use, laboratory systems, patient administration, and local dashboards. A unified event model could permit one patient pathway to contribute appropriately to bacteraemia, device, antimicrobial, SSI, and outbreak surveillance without repeated manual entry. The aim should be semantic interoperability, not merely a larger central database.[7, 41, 42]

The third lesson is to report utilisation as well as harm. A reduction in catheter use is a prevention achievement even before infection rates change. National device-utilisation measures would help identify avoidable exposure and support quality improvement where infection counts are too small for stable comparison. Such process-linked denominators can also make feedback faster because device use is observable in near real time.[27, 45]

England should nevertheless avoid importing US financial incentives without adapting them to the NHS context. Payment penalties can focus attention but may amplify coding disputes and inequity. National oversight can instead combine minimum standards, improvement support, transparent peer comparison, and proportionate regulatory action. The purpose of adopting richer metrics should be to improve causal learning, not to create a more elaborate punitive league table.

7.2 Lessons for the United States

The first English lesson is the value of national organism surveillance that extends beyond device and procedure modules. Bacteraemia programmes can reveal the changing burden of pathogens across hospital and community interfaces, even when no single device or procedure explains the event. The enhanced English distinction between hospital onset and community onset with recent healthcare exposure is especially relevant as care shifts to ambulatory, long-term, and home settings. US national reporting could more prominently integrate longitudinal healthcare exposure rather than relying on the current admission as the principal frame.[10, 21]

The second lesson is that stable, comprehensible counts retain value alongside complex risk-adjusted metrics. SIRs are useful for comparison but can become opaque to clinicians and the public. Displaying observed cases, predicted cases, denominators, and trajectories together would make the analytic model auditable. National organism trends can also detect broad epidemiological change that is difficult to infer from multiple facility-level modules.[9, 45]

The third lesson is to separate learning from payment more clearly. NHSN serves surveillance, quality improvement, research, public reporting, and reimbursement. These functions can pull definitions in different directions. A measure optimised for payment stability may not be the best early-warning signal, while a sensitive outbreak measure may be unsuitable for financial penalty. Creating explicit tiers—a high-sensitivity learning layer, a validated public-comparison layer, and a narrowly governed payment layer—would reduce pressure on one indicator to satisfy incompatible purposes.[7, 13]

8 A convergent model for next-generation surveillance

8.1 Dual-layer detection and adjudication

The most promising synthesis is a dual-layer architecture. The first layer would use automated, high-sensitivity detection from microbiology, antimicrobial orders, device records, vital signs, diagnostic

codes, procedures, imaging reports, patient movement, and readmissions. It would identify possible events rapidly and with consistent computational rules. The second layer would apply clinical adjudication to cases where attribution or preventability matters, using infection-prevention expertise and structured uncertainty. Straightforward laboratory events could flow automatically; ambiguous pneumonia, secondary bloodstream infection, or SSI cases would be prioritised for review.[41, 42, 44]

This arrangement avoids two common errors. The first is assuming that automation eliminates judgment. Algorithms embed definitions, data quality, and thresholds, and their errors may be systematic. The second is requiring expert review of every record, which is unsustainable and diverts staff from prevention. Human review should be concentrated where it changes classification, reveals a pathway, or triggers action. Systems should record why an event was accepted, rejected, or left uncertain, creating a feedback set for algorithm improvement.[7, 35]

8.2 A common event model

A common event model should distinguish five elements: detection, syndrome, organism, attribution, and consequence. Detection records what triggered the event, such as a positive culture or antibiotic escalation. Syndrome records the clinical infection category. Organism records microbiological findings and resistance. Attribution records hospital onset, prior healthcare exposure, device or procedure association, and uncertainty. Consequence records treatment, length of stay, readmission, intensive care, disability, or death. Current systems often collapse several of these into one label, making it difficult to reinterpret cases when definitions change.

The model should support multiple valid views without duplicate data collection. A patient with *E. coli* bacteraemia following catheter-associated urinary infection could contribute to national pathogen burden, a urinary-catheter pathway measure, antimicrobial-resistance surveillance, and local root-cause review. Each output would use the same underlying episode but a different denominator and purpose. Versioned definitions would allow historical data to be reanalysed when rules change, reducing artificial discontinuities.

8.3 Interoperable denominators and longitudinal follow-up

Numerators attract attention, but denominator infrastructure is the foundation of meaningful surveillance. Hospitals need reliable records of patient-days, device-days, device insertion and removal, procedures, risk factors, bed occupancy, staffing, ward movement, and antimicrobial exposure. These data should be captured as a by-product of care rather than manually reconstructed for reporting. Missingness and implausible values should be visible in dashboards and included in facility data-quality scores.[8, 45]

Longitudinal linkage is equally important. Modern episodes cross organisational boundaries: a patient undergoes surgery in one hospital, is discharged, receives antibiotics in primary care, and is readmitted elsewhere. A hospital-only record can miss the infection or assign it to the wrong institution. Privacy-preserving linkage using stable patient identifiers can improve SSI follow-up, detect recent healthcare exposure, and distinguish relapse from new infection. England's national identifiers and integrated NHS structure offer potential advantages, while US health-information exchange and payer claims provide alternative linkage routes. Neither setting has yet achieved routine, comprehensive, clinically validated cross-boundary surveillance.[40–42]

8.4 Validation as a public output

Validation should be treated as part of surveillance rather than an occasional technical audit. National reports should publish the proportion of facilities audited, sampling methods, sensitivity and specificity estimates, common classification errors, timeliness, and completeness of denominator data. When more intensive validation increases recorded infection, reports should explain that ascertainment has improved. England's prevalence-survey validation offers an example of explicitly estimating missed cases; NHSN's recognition that state validation intensity affects SIRs is an important acknowledgement of the same problem.[23, 34]

Automated systems require additional validation. Performance should be assessed across hospitals, EHR products, patient groups, and periods of operational stress. A high overall area under the receiver operating characteristic curve can conceal poor sensitivity in small subgroups or

specific wards. External validation, calibration monitoring, and change control are essential when an algorithm contributes to public reporting or payment. Facilities should be able to inspect the source data and rule path behind each classified event.

8.5 Equity and structural context

HAI metrics rarely place equity at their centre. Patients differ in underlying illness, disability, housing, access to primary care, prior antimicrobial exposure, language, and capacity to manage wounds or devices after discharge. Hospitals differ in capital infrastructure, isolation rooms, staffing, referral roles, and community burden. Risk adjustment can reduce unfair comparison, but it can also normalise preventable inequity if social disadvantage is treated simply as an expected source of infection. The solution is to stratify outcomes and exposures, report structural context, and distinguish adjustment for comparison from analysis for intervention.

A next-generation system should examine whether device use, delayed removal, SSI detection, and infection outcomes vary by age, sex, ethnicity, deprivation, disability, and route of admission, subject to lawful and ethical governance. It should also examine whether surveillance itself is equitable. Post-discharge infections may be more likely to be missed among patients who face barriers to follow-up. Diagnostic testing may differ by ward or patient group. Missingness should therefore be analysed as a potential source of bias, not only a technical inconvenience.

8.6 Learning-oriented accountability

Accountability is necessary because HAI prevention competes with many operational priorities. However, accountability should create learning rather than fear. A balanced national dashboard would combine outcome measures, exposure measures, process reliability, data quality, and improvement trajectory. High rates would trigger structured review and support before punishment, except where organisations fail to report, conceal data, or neglect basic standards. Low rates would not automatically imply excellence if testing and validation were weak.

The distinction between surveillance and investigation should also be explicit. A national

event definition identifies cases consistently; it does not determine individual fault. Local review should examine contributory systems, including workload, device necessity, antimicrobial decisions, environmental conditions, and transitions of care. Feedback should reach the teams able to change those systems. Surveillance that terminates in a board report or payer score without returning to clinical work has not completed its preventive function.[3, 4, 25]

Table 3: Recommended components of a convergent UK-US surveillance model

Component	Operational proposal	Expected benefit
Dual-layer ascertainment	Automated high-sensitivity detection followed by targeted clinical adjudication	Reduces manual workload while preserving clinical meaning
Common event model	Separate detection, syndrome, organism, attribution, and consequence	Allows multiple reports from one episode and transparent reclassification
Exposure denominators	Capture devices, procedures, patient-days, antimicrobial use, and staffing as routine structured data	Links outcomes to modifiable pathways and improves risk adjustment
Longitudinal linkage	Connect acute, primary, outpatient, long-term-care, and readmission records	Improves post-discharge ascertainment and prior-healthcare attribution
Published validation	Report audit coverage, sensitivity, specificity, missingness, and definition changes	Makes data quality visible and reduces rewards for under-detection
Tiered use of measures	Separate learning, public-comparison, and payment indicators	Prevents one metric from serving incompatible purposes
Equity analysis	Stratify exposure, detection, and outcomes while reporting structural context	Identifies unequal risk without normalising it through adjustment
Rapid feedback	Provide ward-level signals with national benchmarking and investigation support	Converts surveillance into timely prevention action

9 Research priorities

The first research priority is comparative validation using a common reference set. Matched hospitals in the United Kingdom and United States could apply English organism definitions, NHSN clinical and LabID definitions, and a blinded expert reference standard to the same de-identified episodes.

This would quantify how much difference arises from definitions rather than epidemiology. It would also reveal which cases are uniquely captured by each system and whether those cases identify distinct prevention opportunities.

Second, research should evaluate the causal effect of reporting and incentives. Time-series improvements following surveillance expansion are encouraging, but they combine many interventions. Natural experiments created by changes in mandates, payment, validation, or public reporting could assess whether effects arise from true prevention, altered testing, or classification. Outcomes should include unreported syndromes, antimicrobial use, mortality, and patient experience to detect displacement or gaming.

Third, automated surveillance should be evaluated prospectively as a service intervention rather than only as an algorithm. Relevant outcomes include time released for infection prevention, speed of cluster detection, agreement with expert review, maintenance cost, adaptability to definition changes, and performance during EHR upgrades or staffing crises. Studies should report source data and implementation maturity in sufficient detail to permit replication.[42, 44]

Fourth, surveillance of non-ventilator hospital-acquired pneumonia requires coordinated development. Given its prevalence, continued exclusion from routine comprehensive surveillance is difficult to justify. A pragmatic programme could combine antibiotic initiation, oxygen change, imaging text, microbiology, aspiration risk, and clinician review, with graded certainty rather than a single binary definition. The objective should be actionable trend detection, not an implausible claim of diagnostic precision.

Fifth, research should identify the best methods for post-discharge SSI ascertainment. Linkage to primary care, emergency, microbiology, pharmacy, and readmission data should be compared with patient-reported outcomes and surgeon review. Algorithms must distinguish wound concern from infection and should estimate sensitivity across social groups. The result could support comparable 30- or 90-day surveillance without imposing universal manual telephone follow-up.[39–41]

A sixth priority is surveillance usability. Researchers should test whether ward teams

understand the measures they receive, whether feedback arrives early enough to alter care, and whether dashboards lead to concrete changes in device removal, antimicrobial review, environmental practice, or escalation. Comparative studies should follow a signal from detection through interpretation, decision, action, and outcome. This would expose the organisational losses that occur when technically accurate data are delayed, stripped of context, or delivered to people without authority to respond locally. Patient representatives should help define which outcomes and explanations make national reporting credible.

Finally, international comparison needs a minimum common dataset rather than a single universal headline rate. A feasible dataset would include admission and discharge dates, specimen and event dates, syndrome, organism, resistance, recent healthcare exposure, device and procedure status, denominator, ascertainment method, validation status, and uncertainty. Countries could retain local systems while producing a harmonised analytic extract. WHO and regional networks are well placed to coordinate such work.[8, 16]

10 Limitations

This review has several limitations. It is narrative and policy oriented, not a systematic review, and does not grade the certainty of every empirical claim. National programmes evolve, and technical manuals may change after publication. The UK discussion necessarily privileges England; surveillance in Scotland, Wales, and Northern Ireland warrants dedicated comparative analysis. The US discussion focuses on NHSN and CMS and therefore gives less attention to Veterans Health Administration systems, private collaboratives, insurer datasets, and local public-health networks.

The current statistics are intentionally descriptive. They arise from different periods, populations, and definitions and cannot establish relative national performance. Even within one system, year-on-year change may reflect case mix, testing, participation, model updates, or validation as well as prevention. Published protocols describe intended practice but cannot fully reveal local implementation. Interviews and ethnographic observation of infection-prevention teams would add

important evidence about how definitions are negotiated in daily work.

The proposed convergent model also creates risks. Greater linkage and automation can centralise sensitive data, increase cyber-security exposure, and produce opaque classifications if governance is weak. More detailed metrics can generate additional reporting burden rather than reduce it. These risks reinforce the need for minimum necessary data, transparent algorithms, strong information governance, and co-design with patients, clinicians, infection preventionists, microbiologists, epidemiologists, and data engineers.

11 Conclusions

The United Kingdom and United States have built sophisticated but differently oriented HAI surveillance systems. England's principal national architecture is comparatively centralised and organism focused, providing long, publicly visible time series for selected bacteraemias and CDI, supplemented by SSI surveillance and point-prevalence surveys. The United States uses NHSN as a broad modular platform for device, procedure, and laboratory events, with risk-adjusted metrics deeply connected to state mandates and federal quality-payment programmes. England is stronger at displaying national pathogen burden across an integrated service; the United States is stronger at exposure-adjusted measurement of defined clinical pathways. Each strength reveals the other's blind spot.

The central implication is methodological and political. Surveillance rates cannot be interpreted apart from the institutions that produce them. Definitions, denominators, testing, validation, post-discharge follow-up, and incentives determine what becomes visible. Direct national comparisons based on headline figures risk mistaking measurement architecture for epidemiology. More useful comparison asks whether each system detects preventable harm promptly, represents the full burden sufficiently, supports action, and treats uncertainty honestly.

Future surveillance should combine automated detection with targeted clinical judgment, pathogen burden with syndrome and pathway measures, hospital episodes with longitudinal

healthcare exposure, and public accountability with validation and improvement support. Such a system would not eliminate disagreement about attribution. It would make that disagreement explicit, structured, and useful. The objective is not a perfect count detached from practice, but a trustworthy learning infrastructure through which patients, professionals, hospitals, and public agencies can recognise preventable infection and act before it recurs.

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The author(s) declare no competing interests.

Ethical approval

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No new dataset was generated. All numerical examples are drawn from the cited public reports.

References

- [1] R. W. Haley, D. H. Culver, J. W. White, W. M. Morgan, T. G. Emori, V. P. Munn, and T. M. Hooton. “The efficacy of infection surveillance and control programs in preventing nosocomial infections in US hospitals”. In: *American Journal of Epidemiology* 121.2 (1985), pp. 182–205. doi: 10.1093/oxfordjournals.aje.a113990.
- [2] C. A. Umscheid, M. D. Mitchell, J. A. Doshi, R. Agarwal, K. Williams, and P. J. Brennan. “Estimating the proportion of healthcare-associated infections that are reasonably preventable and the related mortality and costs”. In: *Infection Control and Hospital Epidemiology* 32.2 (2011), pp. 101–114. doi: 10.1086/657912.
- [3] W. Zingg, A. Holmes, M. Dettenkofer, T. Goetting, F. Secci, L. Clack, B. Allegranzi, A.-P. Magiorakos, and D. Pittet. “Hospital organisation, management, and structure for prevention of health-care-associated infection: a systematic review and expert consensus”. In: *The Lancet Infectious Diseases* 15.2 (2015), pp. 212–224. doi: 10.1016/S1473-3099(14)70854-0.
- [4] J. Storr, A. Twyman, W. Zingg, N. Damani, C. Kilpatrick, J. Reilly, L. Price, M. Egger, M. L. Grayson, E. Kelley, and B. Allegranzi. “Core components for effective infection prevention and control programmes: new WHO evidence-based recommendations”. In: *Antimicrobial Resistance and Infection Control* 6 (2017), p. 6. doi: 10.1186/s13756-016-0149-9.
- [5] T. C. Horan, M. Andrus, and M. A. Dudeck. “CDC/NHSN surveillance definition of health care-associated infection and criteria for specific types of infections in the acute care setting”. In: *American Journal of Infection Control* 36.5 (2008), pp. 309–332. doi: 10.1016/j.ajic.2008.03.002.
- [6] M. J. Durkin, A. W. Baker, K. V. Dicks, S. S. Lewis, R. W. Moehring, L. F. Chen, D. J. Anderson, and D. J. Sexton. “A comparison between National Healthcare Safety Network laboratory-identified event reporting versus traditional surveillance for *Clostridium difficile* infection”. In: *Infection Control and Hospital Epidemiology* 36.2 (2015), pp. 125–131. doi: 10.1017/ice.2014.42.
- [7] M. S. M. van Mourik, E. N. Perencevich, P. Gastmeier, and M. J. M. Bonten. “Designing surveillance of healthcare-associated infections in the era of automation and reporting mandates”. In: *Clinical Infectious Diseases* 66.6 (2018), pp. 970–976. doi: 10.1093/cid/cix835.

- [8] World Health Organization. *Surveillance of health care-associated infections at national and facility levels: practical handbook*. Geneva: World Health Organization, 2024. URL: <https://www.who.int/publications/i/item/9789240101456> (visited on 07/15/2026).
- [9] A. P. Johnson, J. Davies, R. Guy, J. Abernethy, E. Sheridan, A. Pearson, and G. Duckworth. “Mandatory surveillance of methicillin-resistant *Staphylococcus aureus* (MRSA) bacteraemia in England: the first 10 years”. In: *Journal of Antimicrobial Chemotherapy* 67.4 (2012), pp. 802–809. DOI: 10.1093/jac/dkr561.
- [10] UK Health Security Agency. *Mandatory surveillance of Gram-negative bacteraemia, *Staphylococcus aureus* bacteraemia and *Clostridioides difficile* infection: protocol and data definitions*. London: UK Health Security Agency, 2025. URL: <https://www.gov.uk/government/publications/mandatory-surveillance-of-mrsa-bacteraemia> (visited on 07/15/2026).
- [11] Centers for Disease Control and Prevention. *National Healthcare Safety Network patient safety component manual*. Atlanta, GA: Centers for Disease Control and Prevention, 2026. URL: https://www.cdc.gov/nhsn/pdfs/pscmanual/pcsmanual_current.pdf (visited on 07/15/2026).
- [12] Centers for Medicare and Medicaid Services. *Hospital-Acquired Condition Reduction Program*. 2026. URL: <https://www.cms.gov/medicare/quality/value-based-programs/hospital-acquired-conditions> (visited on 07/15/2026).
- [13] L. McKibben, T. Horan, J. I. Tokars, G. Fowler, D. M. Cardo, M. L. Pearson, and P. J. Brennan. “Guidance on public reporting of healthcare-associated infections: recommendations of the Healthcare Infection Control Practices Advisory Committee”. In: *American Journal of Infection Control* 33.4 (2005), pp. 217–226. DOI: 10.1016/j.ajic.2005.04.001.
- [14] T. Hausteijn, P. Gastmeier, A. Holmes, J.-C. Lucet, R. P. Shannon, D. Pittet, and S. Harbarth. “Use of benchmarking and public reporting for infection control in four high-income countries”. In: *The Lancet Infectious Diseases* 11.6 (2011), pp. 471–481. DOI: 10.1016/S1473-3099(10)70315-7.
- [15] S. S. Magill et al. “Changes in prevalence of health care-associated infections in U.S. hospitals”. In: *New England Journal of Medicine* 379.18 (2018), pp. 1732–1744. DOI: 10.1056/NEJMoa1801550.

- [16] C. Suetens, K. Latour, S. Karki, E. Ricchizzi, P. Kinross, M. L. Moro, B. Jans, S. Hopkins, S. Hansen, O. Lyytikainen, J. Reilly, A. Deptula, W. Zingg, D. Plachouras, and D. L. Monnet. “Prevalence of healthcare-associated infections, estimated incidence and composite antimicrobial resistance index in acute care hospitals and long-term care facilities: results from two European point prevalence surveys, 2016 to 2017”. In: *Eurosurveillance* 23.46 (2018), p. 1800516. DOI: 10.2807/1560-7917.ES.2018.23.46.1800516.
- [17] Public Health Scotland. *Healthcare associated infection annual reports*. 2026. URL: <https://publichealthscotland.scot/publications/healthcare-associated-infection-annual-report/> (visited on 07/15/2026).
- [18] Public Health Wales. *Healthcare associated infection surveillance*. 2026. URL: <https://phw.nhs.wales/services-and-teams/harp/healthcare-associated-infections/> (visited on 07/15/2026).
- [19] Public Health Agency Northern Ireland. *Healthcare-associated infection surveillance*. 2026. URL: <https://www.publichealth.hscni.net/directorates/public-health/health-protection/healthcare-associated-infections-antimicrobial-resistance> (visited on 07/15/2026).
- [20] J. Reilly, S. Cairns, S. Fleming, D. Hewitt, R. Lawder, C. Robertson, W. Malcolm, D. Nathwani, C. Williams, L. Cuschieri, I. F. Laurenson, A. Grant, and C. Kilpatrick. “Results from the second Scottish national prevalence survey: the changing epidemiology of healthcare-associated infection in Scotland”. In: *Journal of Hospital Infection* 82.3 (2012), pp. 170–174. DOI: 10.1016/j.jhin.2012.07.024.
- [21] UK Health Security Agency. *Annual epidemiological commentary: mandatory healthcare-associated infection surveillance in England, financial year 2024 to 2025*. London: UK Health Security Agency, 2025. URL: <https://www.gov.uk/government/statistics/mrsa-mssa-and-e-coli-bacteraemia-and-c-difficile-infection-annual-epidemiological-commentary> (visited on 07/15/2026).
- [22] UK Health Security Agency. *Surveillance of surgical site infections in NHS hospitals in England, April 2024 to March 2025*. London: UK Health Security Agency, 2026. URL: <https://www.gov.uk/government/statistics/surgical-site-infections-in-nhs-hospitals-in-england>

uk/government/publications/surgical-site-infections-ssi-surveillance-nhs-hospitals-in-england (visited on 07/15/2026).

[23] Centers for Disease Control and Prevention. *2024 national and state healthcare-associated infections progress report*. Atlanta, GA: Centers for Disease Control and Prevention, 2026. URL: <https://www.cdc.gov/healthcare-associated-infections/php/data/progress-report.html> (visited on 07/15/2026).

[24] D. W. Eyre, M. L. Cule, D. J. Wilson, D. Griffiths, A. Vaughan, L. O'Connor, C. L. C. Ip, T. Golubchik, E. M. Batty, J. M. Finney, D. H. Wyllie, X. Didelot, P. Piazza, R. Bowden, K. E. Dingle, R. M. Harding, D. W. Crook, M. H. Wilcox, T. E. A. Peto, and A. S. Walker. "Diverse sources of *C. difficile* infection identified on whole-genome sequencing". In: *New England Journal of Medicine* 369.13 (2013), pp. 1195–1205. DOI: 10.1056/NEJMoa1216064.

[25] World Health Organization. *Guidelines on core components of infection prevention and control programmes at the national and acute health care facility level*. Geneva: World Health Organization, 2016. URL: <https://www.who.int/publications/i/item/9789241549929> (visited on 07/15/2026).

[26] A. P. R. Wilson, M. Kiernan, H. P. Loveday, A. C. Hayward, J. Morgan, G. Duckworth, B. Cookson, A. Charlett, and D. H. Wyllie. "Recommendations for surveillance priorities for healthcare-associated infections and criteria for their conduct". In: *Journal of Antimicrobial Chemotherapy* 67.suppl 1 (2012), pp. i23–i33. DOI: 10.1093/jac/dks187.

[27] P. Pronovost, D. Needham, S. Berenholtz, D. Sinopoli, H. Chu, S. Cosgrove, B. Sexton, R. Hyzy, R. Welsh, G. Roth, J. Bander, J. Kepros, and C. Goeschel. "An intervention to decrease catheter-related bloodstream infections in the ICU". In: *New England Journal of Medicine* 355.26 (2006), pp. 2725–2732. DOI: 10.1056/NEJMoa061115.

[28] C. Brandt, D. Sohr, M. Behnke, F. Daschner, H. Ruden, and P. Gastmeier. "Reduction of surgical site infection rates associated with active surveillance". In: *Infection Control and Hospital Epidemiology* 27.12 (2006), pp. 1347–1351. DOI: 10.1086/509843.

- [29] P. Gastmeier, D. Sohr, F. Schwab, M. Behnke, I. Zuschneid, C. Brandt, M. Dettenkofer, I. F. Chaberny, H. Ruden, and C. Geffers. “Ten years of KISS: the most important requirements for success”. In: *Journal of Hospital Infection* 70.Suppl 1 (2008), pp. 11–16. DOI: 10.1016/S0195-6701(08)60005-5.
- [30] A. Cassini, D. Plachouras, T. Eckmanns, M. Abu Sin, H.-P. Blank, T. Ducomble, S. Haller, T. Harder, A. Klingeberg, M. Sixtensson, E. Velasco, B. Weiss, P. Kramarz, D. L. Monnet, M. E. Kretzschmar, and C. Suetens. “Burden of six healthcare-associated infections on European population health: estimating incidence-based disability-adjusted life years through a population prevalence-based modelling study”. In: *PLOS Medicine* 13.10 (2016), e1002150. DOI: 10.1371/journal.pmed.1002150.
- [31] E. Zimlichman, D. Henderson, O. Tamir, C. Franz, P. Song, C. K. Yamin, C. Keohane, C. R. Denham, and D. W. Bates. “Health care-associated infections: a meta-analysis of costs and financial impact on the US health care system”. In: *JAMA Internal Medicine* 173.22 (2013), pp. 2039–2046. DOI: 10.1001/jamainternmed.2013.9763.
- [32] P. W. Stone, D. Braccia, and E. Larson. “Systematic review of economic analyses of health care-associated infections”. In: *American Journal of Infection Control* 33.9 (2005), pp. 501–509. DOI: 10.1016/j.ajic.2005.04.246.
- [33] S. S. Magill, J. R. Edwards, W. Bamberg, Z. G. Beldavs, G. Dumyati, M. A. Kainer, R. Lynfield, M. Maloney, L. McAllister-Hollod, J. Nadle, S. M. Ray, D. L. Thompson, L. E. Wilson, and S. K. Fridkin. “Multistate point-prevalence survey of health care-associated infections”. In: *New England Journal of Medicine* 370.13 (2014), pp. 1198–1208. DOI: 10.1056/NEJMoa1306801.
- [34] UK Health Security Agency. *Point prevalence survey of healthcare-associated infections, antimicrobial use and stewardship in England, 2023*. London: UK Health Security Agency, 2025. URL: <https://www.gov.uk/government/publications/healthcare-associated-infections-antimicrobial-use-and-stewardship-point-prevalence-survey-england-2023> (visited on 07/15/2026).
- [35] B. G. Mitchell, L. Hall, K. Halton, D. MacBeth, and A. Gardner. “Time spent by infection control professionals undertaking healthcare-associated infection surveillance: a multi-centred cross-sectional study”. In: *Infection, Disease and Health* 21.1 (2016), pp. 36–40. DOI: 10.1016/j.idh.2016.03.003.

- [36] K. E. Dingle et al. “Effects of control interventions on *Clostridium difficile* infection in England: an observational study”. In: *The Lancet Infectious Diseases* 17.4 (2017), pp. 411–421. DOI: 10.1016/S1473-3099(16)30514-X.
- [37] H. P. Loveday, J. A. Wilson, R. J. Pratt, M. Golsorkhi, A. Tingle, A. Bak, J. Browne, J. Prieto, and M. Wilcox. “epic3: national evidence-based guidelines for preventing healthcare-associated infections in NHS hospitals in England”. In: *Journal of Hospital Infection* 86.Suppl 1 (2014), S1–S70. DOI: 10.1016/S0195-6701(13)60012-2.
- [38] E. K. Tissingh, A. Sudlow, A. Jones, and J. F. Nolan. “Orthopaedic surgical site infection surveillance in NHS England”. In: *Bone and Joint Journal* 99-B.2 (2017), pp. 171–174. DOI: 10.1302/0301-620X.99B2.BJJ-2016-0420.R1.
- [39] R. Troughton, G. Birgand, A. P. Johnson, N. Naylor, M. Gharbi, P. Aylin, S. Hopkins, and A. Holmes. “Mapping national surveillance of surgical site infections in England: needs and priorities”. In: *Journal of Hospital Infection* 100.4 (2018), pp. 378–385. DOI: 10.1016/j.jhin.2018.06.006.
- [40] S. Singh, J. Davies, S. Sabou, R. Shrivastava, S. Reddy, and C. Challand. “Challenges in reporting surgical site infections to the national programme”. In: *Annals of the Royal College of Surgeons of England* 97.6 (2015), pp. 460–465. DOI: 10.1308/rcsann.2015.0027.
- [41] T. P. Quan, R. Hope, J. Clarke, O. B. Spiller, T. E. A. Peto, and A. S. Walker. “Using linked electronic health records to report healthcare-associated infections”. In: *Journal of Antimicrobial Chemotherapy* 73.suppl 2 (2018), pp. ii33–ii39. DOI: 10.1093/jac/dky282.
- [42] J. Cregan, R. Hope, J. Charlesworth, G. Dabrera, S. Hopkins, and J. V. Robotham. “Case ascertainment of a potential centrally implemented, automated system for national surveillance of healthcare-associated infections, England, 2016 to 2023”. In: *Eurosurveillance* 30.42 (2025), p. 2500066. DOI: 10.2807/1560-7917.ES.2025.30.42.2500066.
- [43] P. Zarb, B. Coignard, J. Griskeviciene, A. Muller, V. Vankerckhoven, K. Weist, M. M. Goossens, S. Vaerenberg, S. Hopkins, B. Catry, D. L. Monnet, H. Goossens, and C. Suetens. “The European Centre for Disease Prevention and Control pilot point prevalence survey of healthcare-associated infections and antimicrobial use”. In: *Eurosurveillance* 17.46 (2012), p. 20316. DOI: 10.2807/ese.17.46.20316-en.

- [44] J. D. M. Verberk, S. J. S. Aghdassi, M. Abbas, P. Naucner, S. Gubbels, N. Maldonado, Z. R. Palacios-Baena, A. F. Johansson, P. Gastmeier, M. Behnke, S. M. van Rooden, and M. S. M. van Mourik. “Automated surveillance systems for healthcare-associated infections: results from a European survey and experiences from real-life utilization”. In: *Journal of Hospital Infection* 122 (2022), pp. 35–43. doi: 10.1016/j.jhin.2021.12.021.
- [45] Centers for Disease Control and Prevention. *The NHSN standardized infection ratio: a guide to the SIR*. Atlanta, GA: Centers for Disease Control and Prevention, 2024. URL: <https://www.cdc.gov/nhsn/pdfs/ps-analysis-resources/nhsn-sir-guide.pdf> (visited on 07/15/2026).
- [46] Centers for Medicare and Medicaid Services. *Hospital Inpatient Quality Reporting Program*. 2026. URL: <https://www.cms.gov/medicare/quality/initiatives/hospital-quality-initiative/inpatient-reporting-program> (visited on 07/15/2026).
- [47] M. S. Calderwood, D. J. Anderson, D. W. Bratzler, E. P. Dellinger, S. Garcia-Houchins, L. L. Maragakis, A.-C. Nyquist, K. M. Perkins, M. A. Preas, L. Saiman, J. K. Schaffzin, M. L. Schweizer, D. S. Yokoe, and K. S. Kaye. “Strategies to prevent surgical site infections in acute-care hospitals: 2022 update”. In: *Infection Control and Hospital Epidemiology* 44.5 (2023), pp. 695–720. doi: 10.1017/ice.2023.67.
- [48] L. M. Weiner-Lastinger, V. Pattabiraman, R. Y. Konnor, P. R. Patel, E. Wong, S.-Y. Xu, B. Smith, J. R. Edwards, and M. A. Dudeck. “The impact of coronavirus disease 2019 (COVID-19) on healthcare-associated infections in 2020: a summary of data reported to the National Healthcare Safety Network”. In: *Infection Control and Hospital Epidemiology* 43.1 (2022), pp. 12–25. doi: 10.1017/ice.2021.362.